

120 Days of Quant Finance

Week 2 – Statistical Inference

Day 13: False Discoveries in Financial Research

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Introduction

Financial data is noisy, non-stationary, and highly adaptive. As a result, statistical inference in finance is particularly vulnerable to false discoveries. Many strategies that appear statistically significant in-sample fail to generalize out-of-sample.

This section examines why false discoveries are prevalent in financial research and why naive reliance on statistical significance can lead to misleading conclusions.

What Is a False Discovery?

A false discovery occurs when a researcher concludes that a trading strategy or predictive signal has genuine explanatory or predictive power, when in reality the observed performance is due entirely to random chance.

In hypothesis testing terms, a false discovery corresponds to rejecting the null hypothesis when it is in fact true, i.e., a Type I error.

Why Financial Research Is Especially Vulnerable

Several structural features of financial markets exacerbate the false discovery problem:

- **Low signal-to-noise ratio:** True predictive signals are weak relative to market noise.
- **Massive multiple testing:** Thousands of strategies, indicators, and parameter combinations are evaluated.

- **Non-stationarity:** Market regimes change, invalidating historical relationships.
- **Limited data:** Even long time series contain relatively few independent observations.

Together, these features create an environment where spurious patterns are inevitable.

Multiple Testing and False Positives

Consider testing N independent strategies, each at a significance level α . Even if none of the strategies possess true predictive power, the expected number of false positives is:

$$\mathbb{E}[\text{False Positives}] = N\alpha.$$

For example, testing 100 strategies at $\alpha = 0.05$ yields an expected 5 false discoveries purely by chance.

This phenomenon is known as the **multiple testing problem**.

Limitations of p-values in Research at Scale

The p-value measures the probability of observing data at least as extreme as that observed, assuming the null hypothesis is true.

However, p-values:

- Do not account for the number of tests performed,
- Do not penalize model complexity,
- Do not reflect selection bias in strategy discovery.

As a result, small p-values may arise frequently in large-scale exploratory research, even when no true signal exists.

False Discovery Rate

Rather than controlling the probability of a false positive in an individual test, modern research focuses on controlling the **False Discovery Rate (FDR)**.

Conceptually, the false discovery rate is defined as:

$$\text{FDR} = \mathbb{E} \left[\frac{\text{Number of False Positives}}{\text{Total Number of Rejected Nulls}} \right].$$

FDR reframes inference from evaluating individual hypotheses to assessing the reliability of an entire research process.

Backtest Overfitting and Selection Bias

Backtest overfitting occurs when strategy parameters are optimized to historical noise rather than genuine structure.

Common sources of overfitting include:

- Parameter tuning on the full dataset,
- Repeated re-testing after observing poor results,
- Discarding unsuccessful strategies without accounting for them.

This selection process inflates apparent performance and increases the likelihood of false discoveries.

Why False Strategies Often Look Convincing

False discoveries can produce:

- High Sharpe ratios,
- Smooth equity curves,
- Apparent stability over finite samples.

Random noise, when conditioned on extreme outcomes, can resemble genuine structure, particularly in small or moderate samples.

Defenses Against False Discoveries

There is no single remedy for false discoveries; instead, robust research relies on multiple layers of defense:

Structural Controls

- Limiting degrees of freedom,
- Pre-specifying hypotheses,
- Penalizing model complexity.

Statistical Controls

- Adjustments for multiple testing,
- Higher thresholds for statistical significance,
- Use of deflated or adjusted Sharpe ratios.

Process Controls

- Strict separation of in-sample and out-of-sample data,
- Walk-forward and rolling validation,
- Conservative assumptions about costs and slippage.

Interpretation in Professional Research

In professional settings, backtest results are interpreted with explicit acknowledgment of false discovery risk. Apparent performance is viewed as a hypothesis to be stress-tested rather than as evidence of genuine alpha.

Claims of predictability must survive multiple layers of validation before capital is allocated.

Conclusion

False discoveries are not anomalies in financial research; they are the default outcome in noisy, high-dimensional environments.

Effective quantitative research is therefore less about discovering impressive results and more about systematically eliminating spurious findings. A successful research process prioritizes skepticism, robustness, and humility in the face of uncertainty.